



Mikko De Torres

Mechanical design and automation engineer

SolidWorks Mechanisms Fabrication drawings 3D printing Robotics

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Mechanical design, from concept to a working machine

I design mechanisms and machines in SolidWorks, then build and test them. My work covers the concept, 3D models, drawings, part selection, assembly and testing.

Seven years in industry as a maintenance and automation engineer, then seven teaching robotics, control systems and machine design at Batangas State University.

What I bring to a team

Design it, choose the parts, build it, test it, then document it well enough that someone else can build it again.

Mechanical design and CAD

SolidWorks, Onshape, AutoCAD. Designed the structure and folding mechanism of a DOST-funded solar power station, from fabrication drawings to a deployed unit. Designed the joint drives of a SCARA arm: chain and sprocket, belt and pulley, rack and pinion, and a screw-driven gripper.

Electronics and code

Arduino, ESP32, stepper and servo motors, ROS, Python, MATLAB. I can wire and program what I design.

Education

BS Mechatronics Engineering, 2012. MS Artificial Intelligence, in progress. Both at Batangas State University.

Certifications

TESDA NC II, Mechatronics Servicing.
BOSH Safety Officer 2 (SO2), 2026.



Collapsible solar power station for farms

A portable, foldable solar power unit for farm use, funded by DOST-TAPI and built to a working prototype.

What I built

The mechanical structure and the folding mechanism, modelled in SolidWorks. The unit carries photovoltaic panels and a battery storage unit, and collapses so one team can move, deploy and store it without special equipment. Linear actuators open and close the panel array.



SolidWorks - open



SolidWorks - front

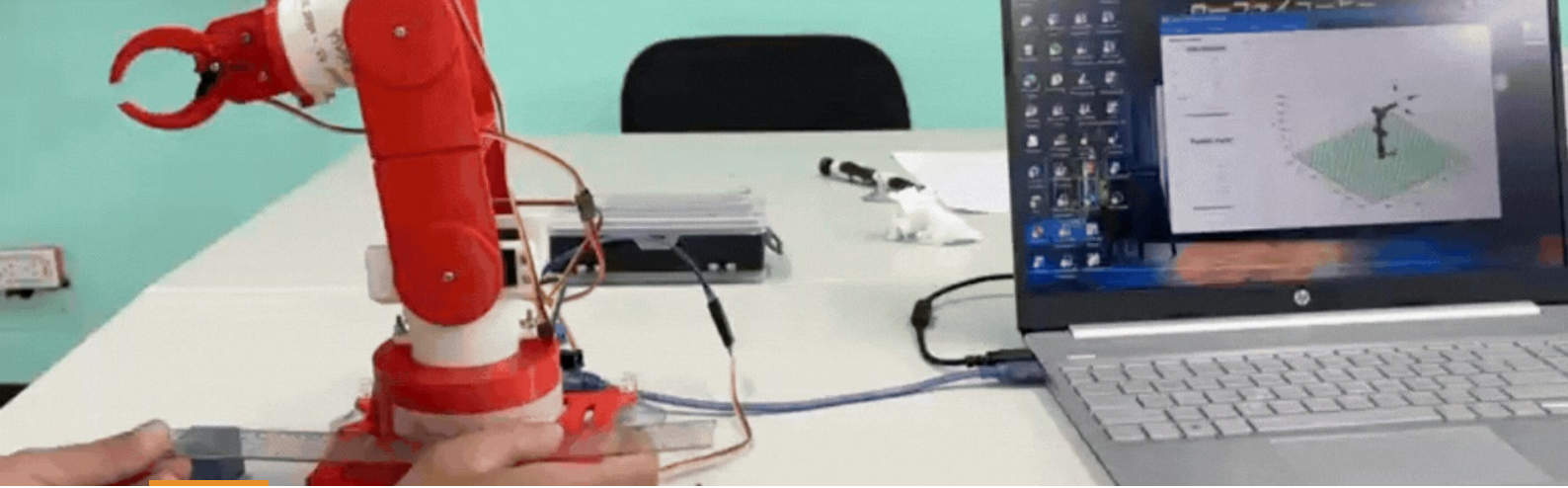


SolidWorks - back

How

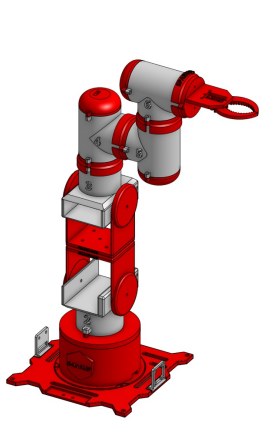
SolidWorks for the structure and mechanism, then fabrication drawings. I also produced the brochure and the instruction materials the users received.

Role Mechanical design lead	Tools SolidWorks, fabrication drawings	Context DOST-TAPI funded project	Year 2023-24
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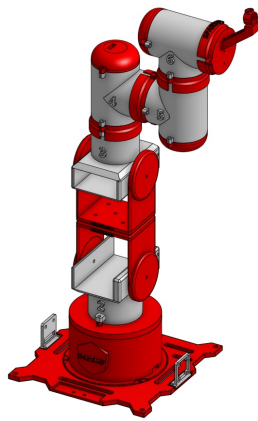


3-DOF to 6-DOF robot arm kit

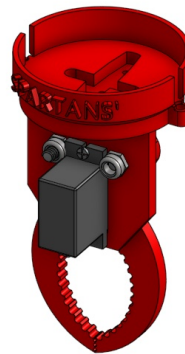
A modular robot arm, 3D printed and assembled from three to six axes depending on the configuration - a teaching platform my students learn kinematics on.



CAD - view 1



CAD - view 2



Gripper design



Assembled arm

What I built

A six-axis arm designed in SolidWorks and 3D printed. Assembled with servo motors, a custom gripper, and an Arduino-based controller. Used in robotics classes for kinematics, pick-and-place and path planning demonstrations.

How

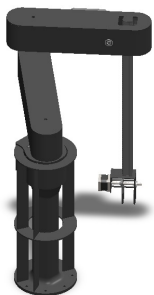
3D printed parts, SolidWorks, Arduino, servo motors. Competed and demonstrated at robotics events.

Role Built and integrated	Tools SolidWorks, 3D printing, Arduino	Context Robotics teaching platform	Year 2023-24
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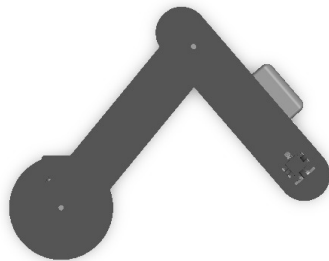


3-DOF RRP SCARA manipulator

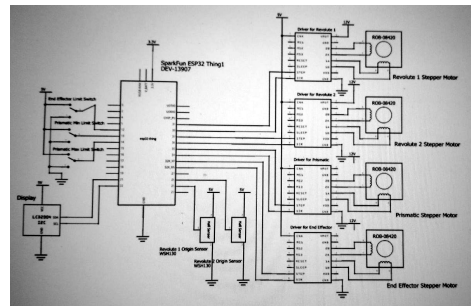
A SCARA-type arm with two rotary joints and a vertical slide. I designed the machine elements and programmed the robot.



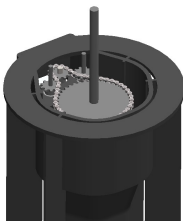
3D model



Top view - reach



Control circuit



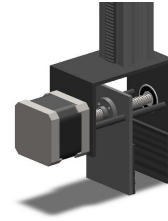
Chain and sprocket



Belt and pulley



Rack and pinion



End effector

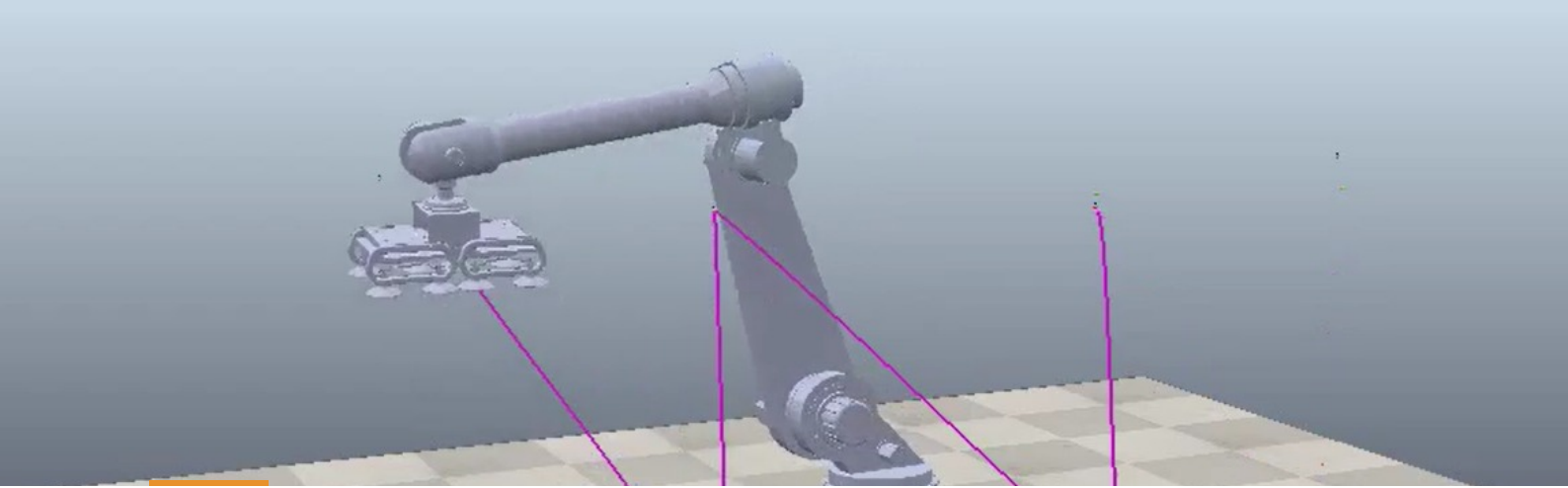
What I built

The drive for each joint: a chain and sprocket in the base, a belt and pulley in the first arm, and a rack and pinion with gear reduction for the vertical slide. The gripper runs on its own stepper motor.

How

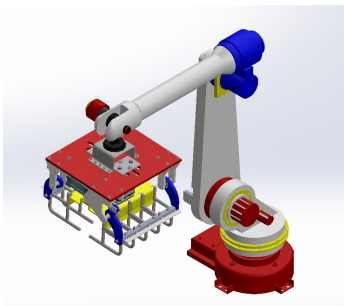
An ESP32 drives four stepper motors. Hall sensors find the home position of the two rotary joints, limit switches stop the slide and the gripper at their ends, and a small LCD shows the readings.

Role	Tools	Context	Year
Design and programming	CAD, ESP32, stepper motors	Robotics Laboratory	2024

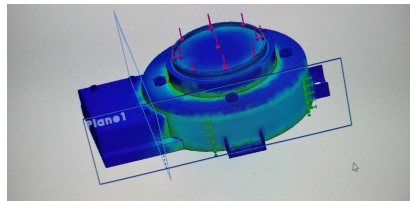


6-DOF manipulator in CoppeliaSim

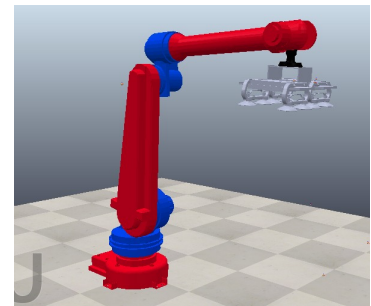
A six-axis arm I designed in SolidWorks, checked for stress, and programmed to move in a CoppeliaSim simulation.



SolidWorks model



Stress study of the base



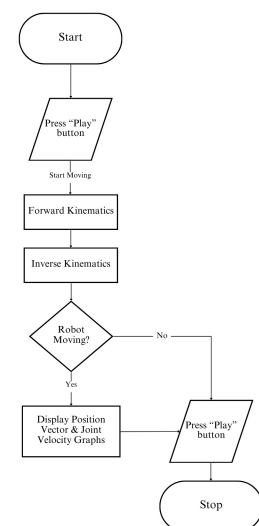
Model in CoppeliaSim

What I built

A six-axis arm modelled in SolidWorks, with a static stress study on the base under load. I brought the model into CoppeliaSim and programmed its forward and inverse kinematics, so the arm follows a set path.

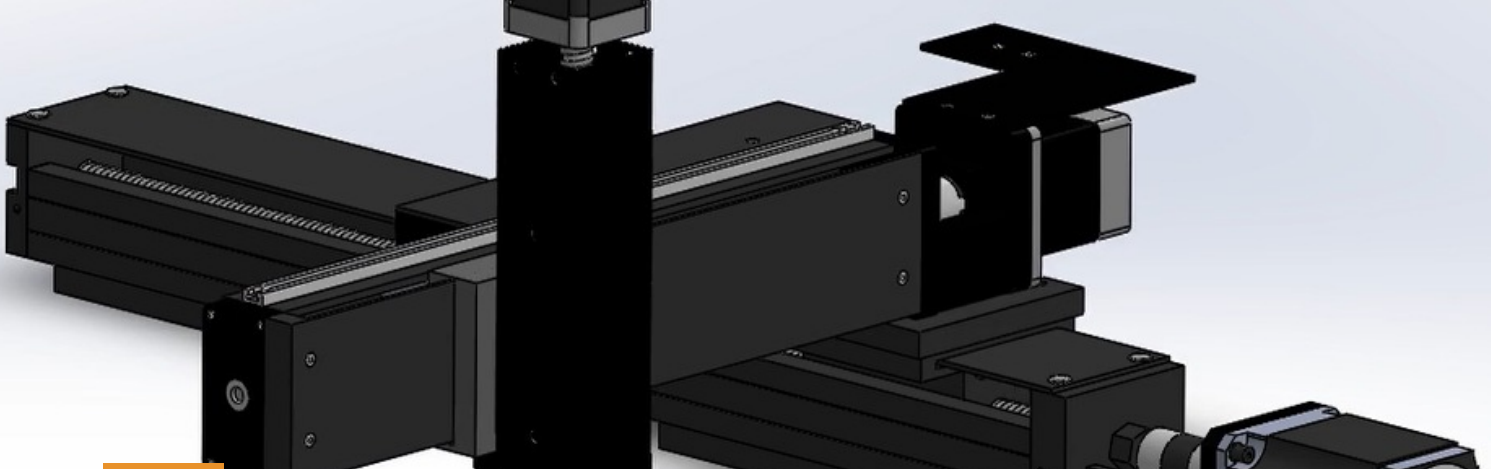
How

Press play and the arm starts moving. While it moves, the simulation plots the tool position and the joint speeds live. SolidWorks for the design and stress study, CoppeliaSim for the simulation.



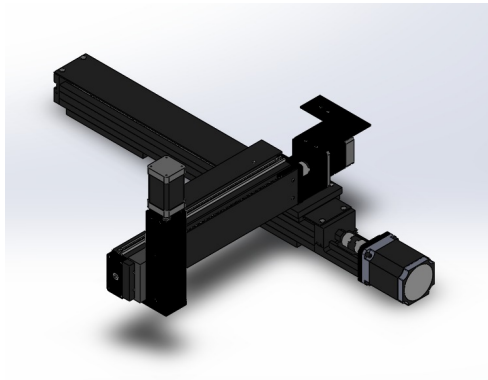
Program flow

Role Design and simulation	Tools SolidWorks, CoppeliaSim	Context Robotics Laboratory	Year 2024
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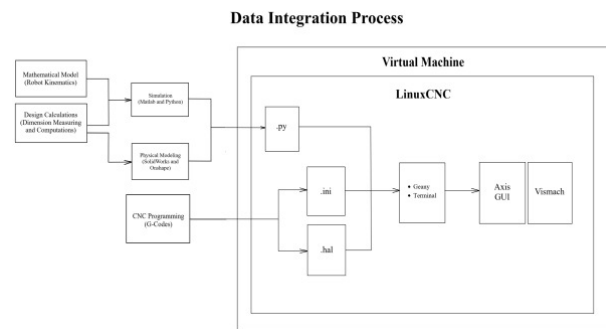


Cartesian CNC robot - digital twin

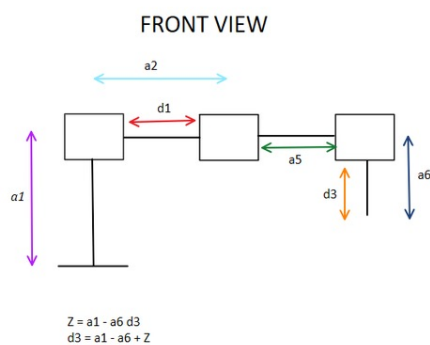
A three-axis Cartesian robot and its digital twin, built with a student team at the Mechatronics and Robotics Laboratory.



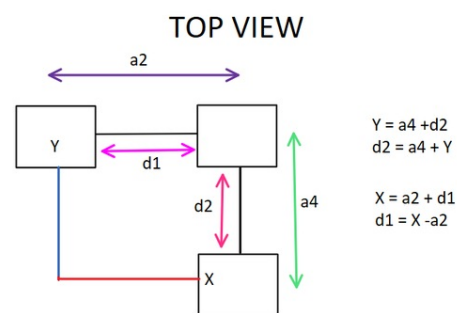
3D model



Data integration



Kinematics - front



Kinematics - top

What I built

The geometric design in SolidWorks and the digital twin, programmed and linked to LinuxCNC. The work covers the kinematics, the data link between the machine and the model, and the engineering drawings.

Role	Tools	Context	Year
Design and programming	SolidWorks, LinuxCNC, MATLAB	Robotics Laboratory	2023